1.CLOUD COMPUTING

Cloud Computing:

Cloud Computing means using computer services through the internet instead of buying and owning them**.**

In cloud computing:

- You don’t need to buy expensive computers or servers
- You don’t need to install software
- you needn't to maintain servers
- You use services online whenever you need them

In simple words:

Cloud computing is like renting technology instead of owning it.

(or)

"Cloud computing means renting computing services like storage, software, and servers through the internet.”

Easy Real-Life Examples for Students

1.**Google Drive**

Instead of using a pen drive, you store files online and access them anytime.

2.**Gmail**

You don’t install email software—emails work through the internet.

3.**Netflix**

You don’t download movies—you watch them online using cloud servers.

4. **Online Classes (Zoom / Google Meet)**

Classes happen through cloud-based platforms.

One-Line Example:

Just like we use electricity without owning a power station, we use cloud computing without owning computers or servers.

This makes cloud computing easy, affordable, and useful for everyone.

What is a Server:

A server is a powerful computer that stores data, runs applications, and provides services to other computers (called clients) through a network or the internet.

or

A server is a computer that works for other computers by storing information and sending it when needed.

Easy Real-Life Example

Think of a restaurant kitchen:

The kitchen = Server

Customers = Users

Food = Data / services

Customers order food, and the kitchen prepares and serves it.

Similarly, users request data, and the server sends it.

Daily Life Examples of Servers

1. Google Server:

When you search on Google, the server:

Finds information

Sends results to your phone or laptop

2. WhatsApp Server:

Stores messages and delivers them to users

3. College Website Server:

Stores: Student data, Results, Notices

Server vs Normal Computer

Normal Computer	Server
Used by one person	Used by many users
Less powerful	Very powerful
Works occasionally	Works 24/7 In Cloud Computing

Traditional computing Vs Cloud computing

Traditional Computing	Cloud Computing
Buy servers & software	Rent services online
High initial cost	Low initial cost
Maintenance required	No maintenance
Limited scalability	Easy to scale
Fixed location	Accessible anywhere

- A retail shop earlier bought billing software and computers, Now it uses cloud-based software online.

Characteristics of Cloud Computing:

Characteristics are the **unique features** that **define cloud computing** and distinguish it from traditional computing.

1.On-Demand Self-Service:

Users can use services without provider interaction.

On-Demand Self-Service means users can use cloud services whenever they need, by themselves, without contacting the service provider or any human support.

The user controls the service through a website or app.

Examples:

- **Google Account Creation:**
  A user can create a Gmail or Google Drive account instantly online without talking to any Google employee.
- **Cloud Storage Increase:**
  A company can increase storage space on Google Drive or OneDrive by clicking a few options online.
- **Virtual Server Creation:**
  A business can create a cloud server on AWS within minutes using an online dashboard.
- **Software Access:**
  A user can start using Zoom or Google Docs immediately after login.

2. Broad Network Access:

Services can be accessed through multiple devices.

Broad Network Access means cloud services can be **accessed over the internet** using **different devices** such as mobiles, laptops, tablets, or desktops.

Users are **not restricted to one location or one device**.

Examples:

- **Zoom Meetings:**

  A student or employee can join the same Zoom meeting using a mobile phone, laptop, or tablet.

- **Google Drive Access:**

  Files stored in Google Drive can be opened from office computer, home laptop, or mobile phone.

- **Gmail**

  Emails can be checked from any device with internet access.

- **Online Banking Apps**

  Customers access banking services using mobile or desktop browsers.

3. Resource Pooling:

Cloud providers pool resources to serve many users.

Resource Pooling means cloud service providers combine computing resources (servers, storage, networks) and share them among many users.

Users do not know or control the exact physical location of the resources, but they get services as required.

Examples:

- **AWS Data Centres**

Many different companies use the same AWS data centre, but their data remains separate and secure.

- **Google Drive Users**

Millions of users store files on Google Drive, all sharing Google's storage infrastructure.

- **Cloud Email Services**

Thousands of businesses use the same email servers provided by Google or Microsoft.

- **Streaming Platforms**

Multiple content creators and viewers use the same cloud infrastructure.

4. Rapid Elasticity:

Resources can expand or shrink quickly.

Rapid Elasticity means cloud resources can be increased or decreased quickly based on user demand, often automatically.

Businesses can scale up during high demand and scale down when demand reduces.

Examples:

Streaming Platforms (Netflix, Hotstar):

During IPL matches or movie releases, the platform increases server capacity to handle millions of users and reduces it later.

E-commerce Websites:
During festival sales, extra servers are added to manage heavy traffic and removed after the sale.

Online Exam Portals:
More resources are added during exam time and reduced afterward.

Food Delivery Apps:
Apps handle more users during peak hours using extra cloud resources.

5. Measured Services:

Uses of services is measured and billed as per usage. Using the resources for rented. Usage is measured and payment is made based on hourly daily or monthly basis for different types of resources.

3. Evolution of Cloud Computing (How It Developed)

Cloud computing did not come suddenly. It developed step by step as technology improved.

1. Traditional Computing (Earlier days):

- Companies bought computers and servers
- Very expensive
- Required maintenance

Example:

A bank having its own data centre in the building

Dis-Advantage: High cost, less flexibility.

2. Client–Server Computing:

- One Central server
- Multiple users (clients) connected
- Data stored in one place
  A college computer lab:
- One main server stores student record.
- All computers access data from it

Improvement: Better data sharing, but still limited and costly.

Example:

College computer lab with one main server

3.Internet & Virtualization:

- Internet became faster
- One physical machine could act like many machines

Example:

One server running multiple websites

Benefit: Reduced cost and better efficiency.

4. Cloud Computing (Present):

- Services available anytime, anywhere
- Scalable and cost-effective

- Services available online
- Access anytime, anywhere
- Scalable (can increase or decrease resources)
- Pay only for what you use
- Handles millions of users during online classes
- Increases servers during peak time
- Reduces usage when demand is low

Result: Flexible, affordable, and powerful.

Example:

Zoom handling millions of users during online classes

Benefits of Cloud Computing:

Cloud computing offers many advantages to organizations:

- **Reduced infrastructure cost:**

No need to purchase servers, networking equipment, or hire maintenance staff, which lowers overall infrastructure expenses.

- **Reduced maintenance cost:**

Hardware, software, power, cooling, and building facilities are managed by the cloud service provider, reducing operational costs.

- **Reduced software cost:**

Organizations do not need to buy or renew individual software licenses, as software is provided on a subscription basis.

- **Automatic software updates:**

Software updates and new versions are automatically handled by the provider, reducing effort and downtime.

- **High availability and reliability:**

Cloud services are available 24/7, and failures are quickly detected and resolved without affecting performance.

- **Improved scalability:**

Resources such as storage and computing power can be increased or decreased easily based on business needs.

- **Dynamic provisioning:**

Additional resources can be quickly allocated during peak workloads and released afterward.

- **Fast execution:**

High processing power enables faster execution of computing-intensive tasks without investing in high-end hardware.

- **Unlimited storage:**

Cloud data centres provide large-scale storage on demand, supporting growing data requirements.

- **Easy maintenance:**

Users are free from managing hardware, software, and storage systems.

- **Access from anywhere:**

Services can be accessed from any location and device, enabling global and round-the-clock operations.

- **Pay-per-use model:**

Users pay only for the resources they use, avoiding underutilization and unnecessary expenses.

- **Rapid deployment:**

Services can be deployed instantly, allowing businesses to focus more on innovation and development.

- **Improved redundancy:**

Data is stored in multiple locations, ensuring data safety during failures or disasters.

- **Greener environment:**

Efficient use of shared resources reduces energy consumption, carbon emissions, and power usage.

<u>Challenges of Cloud Computing:</u>

Despite its benefits, cloud computing has some limitations:

- **Continuous internet connectivity:**

Reliable and high-speed internet is required; service may be affected in poor connectivity areas.

- **Security and privacy issues:**

Risks include data loss, unauthorized access, and confidentiality concerns.

- **Data breaches:**

Storing and transferring data across multiple locations increases the risk of cyberattacks.

- **Governance and regulatory issues:**

Data may be stored in different countries, leading to legal and compliance challenges.

- **Lack of standards:**

Absence of uniform standards makes interoperability difficult and leads to vendor lock-in.

- **Integration issues:**

Integrating existing applications with cloud platforms can be complex.

- **Service availability and vendor lock-in:**

Dependence on a single provider can cause issues during service failures or migration.

CLOUD STANDARDS & REFERENCES:

1. Authentication & Authorization

Authentication:

- Verifies **who the user is** (identity check)

Example*:*
Logging in using username and password or OTP.

Authorization:

✓ Decides **what the user can access** after login

Example:
An employee can view data but cannot delete it.

2. IETF (Internet Engineering Task Force)

Meaning:
IETF develops **internet security protocols** and standards.

Security Technologies by IETF

- **SSL (Secure Sockets Layer)**
  Encrypts data sent between browser and server.
- **TLS (Transport Layer Security)**
  Advanced version of SSL, widely used today.

Example:
"https://" websites use TLS.
Uses digital certificates to verify websites.

Example:
SSL certificates for banking websites.

OAuth (Open Authorization Protocol)
Allows login using third-party accounts.

Example:
"Login with Google / Facebook".

- **XACML (eXtensible Access Control Markup Language)**
  Defines **access control rules**.

Example:
Who can read or edit company data.

- **SAML (Security Assertion Markup Language)**
  Enables **single sign-on (SSO)**.

Example:
One login for multiple cloud applications.

3. OASIS (Organization for the Advancement of Structured Information Standards)

Meaning:
Develops **security and access control standards**.

- **XACML** – Policy-based access control
- **OAuth, SAML** – Secure login & identity sharing

Example*:*
Enterprise cloud access control systems.

4. Automated Password Generator

Meaning:
Automatically creates **strong and complex passwords**.

Example:
Password suggestions during signup.

- Improves security
- Prevents weak passwords

5. NIST (National Institute of Standards and Technology)

Meaning:
US organization providing **security standards & guidelines**.

6. Confidentiality

Meaning:
Ensures data is **accessible only to authorized users**.

Technologies Used

- **SSL / TLS** – Secure data transfer
- **Encryption** – Data protection

Example*:*
Encrypted online banking transactions.

7. Key Management Interoperability Protocol (KMIP)

*(***IETF / OASIS / W3C) WORLD WIDE WEB CONSORTIUM**

Meaning:
Standard for **managing encryption keys** securely.

Example:
Managing keys used to encrypt cloud data.

8. XML Encryption Syntax and Processing (W3C)

World wide web consortium

Meaning:
Standard to **encrypt XML data**.

Example*:*
Secure exchange of business data between systems.

9. Encryption Standards (NIST)

- **EES (Escrowed Encryption Standard)** – Older encryption method
- **AES (Advanced Encryption Standard)** – Widely used secure encryption

Example:
AES is used to protect files and databases in cloud.

10. Security Monitoring & Incident Management

SCAP (Security Content Automation Protocol)

*(***NIST)**

Meaning:
Automates **security checks, vulnerability detection**, and compliance.

Example:
Finding system weaknesses automatically.

11. ITU-T (International Telecommunication Union – Telecom)

Meaning:
Develops **global security and communication standards**.

Example:
Secure international data exchange standards.

12. Security Policy Management

XACML (OASIS)

Meaning:
Defines **who can access what data and under which conditions**.

Example:
Role-based access in cloud systems.

13. Incident Management Guidelines (ISO)

Meaning:
Provides **guidelines to handle security incidents**.

Example:
Steps to follow after a data breach.

14. Availability & Continuity Management

Meaning:
Ensures cloud services are **always available**, even during failures.

Example:
Backup servers during system failure.

15. Open Cloud Computing Interface (OCCI)

Meaning:
Standard for **managing cloud services**.

Example:
Controlling cloud resources across platforms.

16. Interoperability

Meaning:
Ability of **different cloud systems to work together**.

Standards

- **CDMI** – Cloud data management
- **CPIP** – Portability profiles
- **SIIF** – Intercloud federation

Example:
Moving data between different cloud providers.

17. Data Portability (SNIA)

Meaning:
Ability to **move data easily** from one cloud to another.

Example:
Migrating files from AWS to Azure.

18. System Portability

OVF (Open Virtualization Format)

Meaning:
Standard for **moving virtual machines** across platforms.

Example:
Running the same VM on different cloud providers.

<u>Summary:</u>

Cloud security and interoperability standards ensure authentication, confidentiality, availability, and smooth data portability across cloud platforms

Chapter 2: Cloud Deployment Models and Cloud Service Delivery Models

Introduction

Cloud computing allows businesses to use computing resources such as **servers, storage, software, and networks** through the internet.
To use cloud services effectively, organizations must understand:

1. **How the cloud is deployed** → *Deployment Models*
2. **What type of service is delivered** → *Service Delivery Models*

Cloud Deployment Models:

Meaning of Cloud Deployment Model:

A **cloud deployment model** defines **where the cloud infrastructure is located**, **who owns it**, and **who can access it and management of cloud services**.

We have four 4 Different Cloud Models:

1.Public Cloud

2.Private Cloud

3.Hybrid Cloud

4.Community Cloud

Most of the advantages and challenges what we have discussed in chapter 1 applicable to these models.

1. Public Cloud:

Definition:

A **public cloud** is owned and managed by a **third-party cloud service provider**. Services are offered to the public over the internet.

Public cloud offers application, storage, computing, networking and other services over the internet by the cloud service providers to the public.

General public, enterprise, government, individuals, industry groups use pubic cloud.

Key Features:

- Shared infrastructure
- Pay-as-you-use pricing
- Accessible from anywhere

Examples:

- Amazon Web Services (AWS)-EC2, S3- simple storage services
- Microsoft Azure
- Google Cloud

Few basic services are providing free of cost and more advances are charged as pay-per-use model

e- mail services, social networking (FB, Twitter)

Cloud storage (Google drive, one drive, Drop Box)

Office Suite (Google Docs, MS 365)

Advantages:

- Low cost
- No hardware maintenance
- Highly scalable

Disadvantages:

- Security concerns
- Less control over data

Benefits:

- **Flexibility and elasticity**
- **Self Service**
- **Availability and reliability**

Example:

A startup uses **Google Drive** to store files instead of buying servers.

Business Reason: Low cost, quick launch, easy set up

2. Private Cloud:

Definition:

A **private cloud** is used by **one organization only,** to make services accessible for the designated organization only. It may be hosted internally or by a third party.

Key Features

- Dedicated infrastructure (The infrastructure is located within the organization itself internally or externally dedicated to is organization employees and stake holders)
- High security
- Full control

Advantages:

- Better data security (big advantage organization has full control over its data)
- Security guidelines
- System performance
- Customized to business needs

Disadvantages:

- High cost
- Requires skilled IT staff

Some of the popular private clouds are Amazon VPC (Virtual Private Cloud), IBM (International business machine) Smart Cloud, Microsoft Private Cloud.

There are two Variants of Private cloud depending on the location of infrastructure

1. On site private cloud
2. 2. Outsourced Private cloud

Example:

A bank uses a private cloud to store **customer financial data**.

Business reason: High Security, full control

3. Hybrid Cloud:

Definition:

A **hybrid cloud** combines **public and private clouds from different service providers**.

It allows an organization to use the public for scalability and low cost. While keeping important and sensitive data inside its own private cloud.

allowing data and applications to move between them.

Key Features

- Flexible
- Balanced security and cost

Advantages

- Sensitive data kept private inside corporate network
- Cost-effective and scalability
- flexible

Disadvantages

- Complex management – managing multiple security system is difficult
- Integration issues
- Communication between cloud is complex

<u>Example or where hybrid cloud is best:</u>

1.A company stores customer data in a private cloud and runs applications on a public cloud.

Ex: e-commerce sales during festival season customer data private and public cloud is used for campaigning)

2.Using SaaS with Secure Data

- An organization wants to use SaaS from public cloud
- But sensitive data should stay in private cloud
- Both clouds are securely connected

Business reason:

Balance between cost and security

<u>4. Community Cloud:</u>

Definition

A **community cloud** is shared by **several organizations** with common goals or requirements and shared interest such as security, business domain and compliance.

Key Features

- Shared cost
- Shared policies

Advantages

- Cost sharing
- Better collaboration

Disadvantages

- Limited scalability
- Shared responsibility

Example

Several universities share a cloud platform for research purposes having common compliance and share resources.

Google Apps for Government and Microsoft Government Community Cloud

Differences Between Public Cloud and Private Cloud

Feature	Public Cloud	Private Cloud
Access	Public who subscribes	Clients and stakeholders of the organization
Connectivity	Internet	Internet, fibre optics, private networks
Ownership & Control	Service provider	Organization (private ownership)
Security	Medium to low	Very high
Scalability	Very high	Limited
Privacy	Limited	Very high
Reliability	Limited	Very high
Cost	Low, pay-as-you-use, shared cost	High, fixed cost
Resources	Shared	Dedicated
Location	Public / off-campus	On-site or dedicated off-campus
Flexibility	High	Limited
Customization	Limited	High

Feature	Public Cloud	Private Cloud
Performance	Depends on Internet (lower)	High, as per requirements
Suitability	Non-mission-critical apps	Mission-critical apps
Tenancy	Multi-tenancy	Single tenancy

Summary of Deployment Models:

Model	Ownership	Users	Cost	Security
Public	Provider	Public	Low	Medium
Private	Organization	Single	High	High
Hybrid	Both	Mixed	Medium	High
Community	Group	Community	Medium	Medium

Part B: Cloud Service Delivery Models

Meaning of Cloud Service Model

A **cloud service model** defines **what type of service is provided to the user**.

Services = What you use online instead of owning on your computer

What are "Services" in Cloud Computing?

Cloud services include:

1. Storage Services:

storing data online instead of storing in computer or physical hard disk

Used to store data online.

Example:

Google Drive, Dropbox – store files, photos, documents.

2. Software Services:

Used to run applications without installing them through the internet using browser.

Example:

Gmail, Microsoft Office 365 – use software through a browser.

3. Computing Services:

Used to run programs and websites.

Example:

AWS or Azure provides servers to host websites and apps.

4. Database Services:

Used to store and manage large data.

Example:

Online banking systems storing customers data

These services include:

Storage – saving data online

Software – using applications online

Servers – powerful computers that run applications

Databases – storing and managing information

Networking – connecting systems

All these are provided online and users pay only for what they use.

Simple Example for Understanding

Think of cloud computing like using a mobile recharge plan:

You don't buy a mobile tower

You pay only for the data you use

Business Example (Startup Company)

A startup company:

Does not buy costly servers

Uses Amazon Web Services (AWS) to run its website and apps

Pays monthly or hourly based on usage

Example:

An online shopping startup uses AWS to store customer data and run its website. During sales, it uses more services and pays more. On normal days, it pays less.

Cloud Service models:

The fundamental concept of cloud computing is to deliver the expected software services

Cloud Computing provides 3 major cloud services models:

1.IaaS

2.Paas

3.SaaS

1. Infrastructure as a Service (IaaS):

Definition:

Provides basic computing infrastructure like servers, storage, and networking

Infrastructure as a Service (IaaS) is a cloud computing model that provides **basic computing infrastructure** over the internet. It includes **servers, storage, networking, operating systems, and security** needed to develop and run applications.

IaaS is the **most basic and fundamental layer** of cloud services. Instead of buying physical hardware, users can **rent computing resources** such as:

- Physical or virtual machines
- Storage devices
- Network resources
- Firewalls and load balancers

Key Points of IaaS:

- Resources are provided **on demand**
- Based on **utility computing** (pay only for what you use)
- Users can deploy:
    - Applications
    - Databases
    - Development tools
- Service provider manages the **hardware**
- Users manage **OS, applications, and data**

Simple Example:

IaaS is like **renting an empty house**:

- Owner provides the building (infrastructure)
- You decide how to use it (applications, software)

User Controls

- Operating system
- Applications
- Data

1. <u>Operating System (OS) Control:</u>

If the user has control over the **operating system**, they can:

- Install or update the OS (Windows, Linux)
- Configure system settings
- Manage security patches

<u>Example:</u>
In **IaaS**, the user chooses and manages the OS.

Here we can use like OS like Linux or Windows based on us

2. <u>Applications Control:</u>

If the user controls **applications**, they can:

- Install software
- Develop and modify applications
- Update or remove applications

Example:
In **IaaS and PaaS**, users manage their applications.

3. <u>Data Control:</u>

If the user controls **data**, they can:

- Upload and delete data
- Decide data access permissions
- Secure and back up data

Example:
In **all cloud models (IaaS, PaaS, SaaS)**, data is controlled by the user.

Advantages:

- High flexibility

Simple words:
High flexibility means you can increase, decrease, or modify resources whenever required.

- Pay only for resources used

Examples:

- Amazon EC2
- Microsoft Azure Virtual Machines

Example:

A company rents virtual servers instead of buying physical servers.

Characteristics of IaaS (Infrastructure as a Service):

1. Virtualization:

- Virtualization is the **core concept of IaaS**.
- It allows **multiple users (multitenancy)** to share resources.
- Ensures **resource pooling and efficient utilization**.

2. Cloud Bursting

- When local resources are insufficient, extra workload is **shifted to the cloud**.
- Used during **peak demand**.
- Helps in **cost reduction**.

3. Geographically Distributed Data Centres

- Data centres are located **across the world**.
- Improves **availability, reliability, and response time**.

4. Advance Reservation of Resources

- Users can **reserve servers or storage in advance**.
- Useful for **planned workloads**.

5. Elasticity and Scalability

- Resources can be **scaled up or down dynamically**.
- Meets changing business needs efficiently.

<u>Benefits of IaaS:</u>

1. Greater Control for Consumers

- Users control:
    - Operating system
    - Applications
    - Databases and tools
- They can **upgrade and manage software** as needed.

2. No Need to Buy Expensive Hardware

- High-end software can run on **powerful cloud servers**.
- No need to purchase costly physical servers.

3. Ideal for Development and Testing

- Developers can test applications on **multiple environments**.
- Useful for testing:
    - Operating systems
    - Device drivers
- No risk to local hardware.

4. Reduced Management Overhead

- Service provider handles:
    - Failover
    - Security
    - Backup and recovery
    - Software updates

<u>Limitations of IaaS:</u>

1. Network and Browser Dependency

- Like SaaS and PaaS, IaaS depends on:
    - Internet connectivity
    - Browser security

2. Security Vulnerabilities

- Users may run **old (legacy) software**.
- Such software can introduce **security risks**.

3. Virtual Machine Sprawl

- Users may create many virtual machines.
- Some VMs may become **outdated or unsecured**. Service provider automatically updates such virtual machines.

4. Data Erase Issues

- Storage is **shared** among users.
- When resources are released, provider must ensure:
    - **No data residue** remains for next user.

5. Legal and Compliance Issues

- Not suitable where:
    - Government data laws
    - Data storage and transmission rules are strict and not compliant.

2. Platform as a Service (PaaS):

Definition

Platform as a Service (PaaS) is a cloud computing model that provides a complete development platform over the internet. It includes resources such as operating systems, programming languages, compilers, runtime environments, databases, and web servers.

PaaS is mainly used by application developers to develop, test, and deploy applications without worrying about purchasing, configuring, or managing the underlying hardware and software platform.

Examples of PaaS tools and resources include database management systems, enterprise service buses, and business intelligence tools.

Simple:

PaaS provides a cloud-based platform that allows developers to build, test, and deploy applications without managing the underlying infrastructure.

PaaS provides a **platform** to develop, test, and deploy applications.

User Controls

- Application development
- No need to manage hardware or OS

Advantages

- Faster development
- Reduced complexity

Examples

- Windows Azure: SQL, python
- Amazon: DB, S3
- Google App Engine
- Heroku

Example

A developer builds a mobile app using Google App Engine without worrying about servers.

Platform as a Service (PaaS) provides a ready-made platform where developers can build, test, and deploy applications without managing hardware or operating systems.

1. <u>Google App Engine</u>

What it is

Google App Engine is a **PaaS offered by Google Cloud** that allows developers to create and run applications on Google's infrastructure.

Example:

A startup develops a **food delivery app** using Google App Engine without worrying about server management.

2. <u>Heroku</u>

What it is

Heroku is a **cloud-based PaaS platform** used to deploy and manage web applications.

Example:

A student creates a **college management web app** and deploys it on Heroku easily.

<u>Basic Characteristics of PaaS:</u>

1. Browser-Based Development Environment

- PaaS provides **web/browser-based tools**.
- Multiple users can **work together at the same time**.
- Developers can:
    - Design
    - Develop
    - Test
    - Integrate
    - Deploy applications
- Most PaaS platforms offer a **Web-based GUI** for easy use.

2. Multiuser and Multitenant Support

- PaaS supports **many users simultaneously**.
- Uses **multitenant architecture**.
- Ensures **scalability** for multiple concurrent users.

3. Common Services and Packages

- Provides ready-to-use services such as:
    - Databases
    - Networking
    - Payment integration
- Saves development time and effort.

4. Built-in Security and Scalability

- PaaS includes **built-in security features**.
- Automatically handles **scaling of applications**.
- Provides **web service interfaces (APIs)** to connect with other applications outside the PaaS platform.

Benefits of PaaS:

1. Control Over Application Development

- Developers have full control over developing and upgrading applications.
- They can modify the application whenever needed**.**

2. Reduced Maintenance Work

- The service provider takes care of:
    - Platform maintenance
    - Tool updates
    - System upgrades
- This reduces administrative effort for developers.

3. Cost Effective

- Payment is made only for the time and resources used.
- No need to buy or set up a platform for short-term use.
- Helps in reducing overall cost.

4. Multi-Platform Support

- Developers can work on multiple platforms using a single PaaS environment.
- No need to own different platforms separately.

5. Automatic Resource Management

- Resource needs (CPU, memory, storage) are handled automatically by PaaS.
- Applications scale up or down based on demand.

Limitations of PaaS:

1. Browser and Connectivity Risks:

- Like SaaS, PaaS is accessed via the Internet.
- Risks include:
    - Browser-based attacks
    - Network reliability issues
    -

2. Interoperability and Portability Issues:

- Different PaaS providers may use different interfaces.
- Moving applications from one provider to another is difficult.
- Example: File handling or database interfaces may differ.

3. Security Concerns:

- Data travels over the Internet.
- Applications must use:
    - Encryption
    - Secure connections
- Security is partly the developer's responsibility.

4. Scheduling Constraints

- PaaS applications are often event-driven.
- They must respond within fixed time limits.
- Resource usage is restricted by the provider.

3. Software as a Service (SaaS):

Definition:

SaaS is software distribution model in which applications are hosted by vendor or service provider and made available to customers over a network i.e. internet.

SaaS provides **ready-to-use software** over the internet,

In SaaS provider manages complete set of resources for SaaS provision. As a consumer we are not aware of the technology and infrastructure details behind successful service consumptions.

User Controls

- Only application usage

Advantages

- Easy to use
- No installation required

Examples

- E-mail (Gmail, Yahoo)
- Social networks (Facebook, linked In, Twitter, YouTube)
- Security (McAfee)
- Zoom
- CRM and HR (Salesforce)
- Backup and recover

Example

Using **Gmail** for email instead of installing email software.

<u>Characteristics of SaaS (Software as a Service)</u>

1. **On-demand and Internet-based**
    - SaaS applications are available **whenever needed**.
    - They are accessed through the **Internet** using a browser.
2. **Scalable services**
    - Users can **increase or decrease usage** easily.
    - No need to buy extra hardware or software.
3. **No back-end knowledge required**
    - Users do **not need to know** how servers or databases work.
    - Everything is handled by the service provider.
4. **Managed by SaaS provider**
    - The provider manages the **entire application stack** (software, server, storage, network).
5. **Multitenant model**
    - A **single software instance** serves **multiple users/organizations**.
    - Each user's data is kept separate and secure.
6. **Automatic updates**
    - Software, technology, and licenses are **updated automatically**.
    - Users always get the **latest version**.

7. **Pay-as-you-go model**
    - Users pay **only for what they use**.
    - Usually based on **monthly or yearly subscription**.

<u>Benefits of SaaS:</u>

1. **Reduced cost and effort**
    - No need to develop or install complex software.
    - Saves **time, money, and manpower**.
2. **Easy software updates**
    - Software is hosted in a **central cloud location**.
    - Updating one cloud version updates it for all users instantly.
3. **Pay only for required modules**
    - Organizations can subscribe to **only needed features**.
    - Example: In SAP, a company may choose only **inventory or finance modules**.
4. **No licensing issues**
    - Software licenses are managed by the provider.
    - Licenses are **automatically updated**.
    - Floating licenses can be shared among users.
5. **Bug fixing handled by provider**
    - Identifying and fixing software errors is the **provider's responsibility**.
6. **Automatic technology upgrades**
    - Users always work with **latest technology** without extra cost.
7. **Access through thin clients**
    - SaaS can be used on **low-end devices**.
    - No need to install software on personal computers.
8. **Centralized data management**
    - Data backup, recovery, security, and power management are handled centrally.

- Reduces risk of **data loss**.

Limitations of SaaS:

1. Data Stored with Provider

- Customer data is stored on the **service provider's servers**.
- Strong security is required to protect data.
- Sometimes **third-party security services** are needed.

2. Limited Customization

- SaaS software is **common for all users**.
- Organizations cannot easily **modify or customize** the software as per their needs.

3. Integration Issues

- SaaS applications may **not integrate easily** with existing systems.
- Especially difficult when combining with older or private cloud applications.

4. Browser Dependency and Security Risk

- SaaS is mostly accessed using **web browsers**.
- Browsers can be affected by **malware or cyber-attacks**.

Situations Where SaaS Is Not a Good Choice

1. Country-Specific Data Laws

- Some countries do not allow data to be stored **outside their region**.
- SaaS providers may store data in foreign data centres.

2. Real-Time, High-Speed Processing

- Applications that need **very fast, real-time data processing** may not work well.
- Internet delays can affect performance.

SaaS Limitations – Simple

- Data security → **Risk**
- Customization → **Limited**

- Integration → **Difficult**
- Browser dependency → **Vulnerable**

SaaS Not Suitable – simple

- Country-specific data laws → **Compliance**
- Real-time fast processing → **Latency**

Cloud Service Models (IaaS, PaaS, SaaS)

Use a real-life analogy (restaurant or housing).

Service Models Explained Simply

Model	What the Business Gets	Example
IaaS	Infrastructure on rent	AWS EC2
PaaS	Platform to build app s	Google App Engine
SaaS	Ready-to-use software	Gmail, Salesforce

Business Perspective:

IaaS → More control, more responsibility

PaaS → Faster development

SaaS → No maintenance, easy to use

<u>Difference Between Deployment and Service Models</u>

Deployment Model	**Service Model**
Defines *where* cloud is located	Defines *what* service is provided
Focus on ownership & access	Focus on service level
Example: Public cloud	Example: SaaS

<u>SUMMARY:</u>

- **Public** → Shared, low cost
- **Private** → Secure, costly
- **Hybrid** → Flexible

- **IaaS** → Infrastructure on rent
- **PaaS** → Development platform
- **SaaS** → Ready software

<u>What is a VPC?</u>

- A VPC is a **logically isolated section** of a public cloud.
- It behaves like a **private cloud**, but is actually **inside the public cloud**.
- It is created **on demand**.

How Isolation is Achieved in VPC

Security and privacy are ensured using:

- Private IP addresses
- Virtual LAN (VLAN)
- Encrypted communication
- VPN (Virtual Private Network)
- Authentication and encryption

This ensures that **other public cloud users cannot access your data**.

<u>Why It Is Called "Virtual"</u>

- It is not a physical private cloud.
- It is **created virtually using software**.
- Set up **only when required**.

<u>VPC with Private Cloud</u>

- Organizations with a **private cloud** can extend it to a **VPC**.
- Useful during **peak demand or short-term scalability needs**.
- Still maintains **private cloud security benefits**.

<u>Ownership and Management</u>

- Even though it behaves like a private cloud:
    - It is **owned and managed by the public cloud provider**
- Example: **Amazon VPC**

Working of VPC

1. Organization creates a **VPC inside public cloud**
2. Employees access it using **VPN**
3. Sensitive data stays secure inside VPC
4. Other users access the internet normally

1. Case Study: Bank of America Turns to Microsoft Cloud for Digital Transformation

Introduction

Bank of America is one of the **largest financial institutions in the world**. With millions of customers, the bank needed a **secure, scalable, and modern IT infrastructure** to support digital banking services.

To achieve this, Bank of America partnered with **Microsoft Cloud (Azure)** to drive its **digital transformation**.

Problem Faced by Bank of America

- Traditional IT systems were **slow and costly**
- Difficulty in handling **huge customer data**
- Need for **strong security and compliance**
- Rising demand for **online and mobile banking**
- Limited flexibility in scaling IT resources

Solution: Microsoft Cloud (Azure)

Bank of America adopted **Microsoft Azure cloud services** to modernize its operations.

Key cloud solutions used:

- Cloud-based data storage
- Advanced analytics and AI
- Secure cloud infrastructure
- Disaster recovery and backup

Benefits Achieved:

- **Improved security**: Strong encryption and compliance

- **Scalability**: Easily handles millions of transactions
- **Cost efficiency**: Reduced hardware and maintenance costs
- **Faster innovation**: New digital services launched quickly
- **Better customer experience**: Faster and reliable online banking

Conclusion:

By adopting **Microsoft Cloud**, Bank of America successfully transformed into a **technology-driven digital bank**, improving efficiency, security, and customer satisfaction.

2. Case Study: Siemens IT Solutions and Services

Introduction:

Siemens IT Solutions and Services (SIS) is the **IT service arm of Siemens**, a global technology company. Siemens required a **robust cloud infrastructure** to support its worldwide operations and clients.

Challenges Faced by Siemens

- Managing **global IT operations**
- High infrastructure and maintenance costs
- Need for **fast deployment of IT services**
- Supporting **multiple industries and clients**
- Data management across countries

Cloud Solution Adopted:

Siemens adopted **cloud computing and virtualization technologies** to provide IT solutions efficiently.

Cloud services used include:

- Infrastructure as a Service (IaaS)
- Private and hybrid cloud models
- Cloud-based enterprise applications
- Centralized data management systems

Benefits to Siemens

- **Cost reduction** through shared cloud resources

- **Improved efficiency** in IT service delivery
- **Faster deployment** of applications
- **High availability** and reliability
- **Better collaboration** across global teams

Business Impact

- Siemens improved **operational efficiency**
- Enhanced customer satisfaction
- Enabled innovation in industries like healthcare, manufacturing, and energy
- Strengthened its position as a **global IT solutions provider**

Conclusion:

Siemens IT Solutions and Services used **cloud computing** to streamline IT operations, reduce costs, and deliver scalable IT services worldwide.